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The NASA Research Announcement NRA-01-OBPR-05 solicits several coordinated efforts that address critical aspects of the radiation shielding program to protect humans during extended exposures to the natural cosmic radiation and solar energetic particles in deep space. This project addresses two specific elements of the NRA:
1) A Definition study of the Deep Space Test-Bed (DSTB) to be developed for testing transport codes, radiation shielding materials and radiation monitoring devices in the deep space radiation environment. 2) Development of a Charged Particle Spectrometer (CPS) to measure composition, spectra and radiation dose. The CPS will be developed for validation of the accuracy of radiation transport codes and to serve as a radiation-environment monitoring instrument for the DSTB.

The first element of this project is a study to define a balloon-borne DSTB to enable: testing and evaluating proposed space radiation shielding materials, testing and development of new radiation monitoring instrumentation and the

validation of radiation transport codes. The definition study has generated detailed test-bed requirements based on solicited input from the scientific community, provided documentation of the environmental parameters to be monitored during balloon-borne tests and developed a plan for implementing the DSTB facility. The second element of this project describes instrumentation that would utilize the DSTB to measure the elemental spectra of charged particles. This instrumentation will measure the incident Galactic Cosmic Rays (GCR) flux and secondary particle spectra behind shielding materials. These measurements will be used to determine the accuracy of radiation transport codes.

Provide definition and design for the DSTB facility and operations. Develop and demonstrate a Charged Particle Spectrometer for balloon flight exposures to measure the primary GCR and secondary particles produced in candidate radiation shielding materials.

Task Description:

For human exploration of space, protecting astronauts from ionizing radiation hazards must be addressed in spacecraft design to meet NASA's standards for safety, mission success and the Radiation Health Program's goal to keep crew exposures to radiation "as low as reasonably achievable" (ALARA). The principal ionizing radiation of concern in deep space for a manned mission is the "galactic cosmic radiation-GCR" which consists of the nuclei of all the elements, with mean energy of about 5 billion electron volts (5 GeV). Shielding against GCR largely involves nuclear rather than atomic processes. The nuclear interactions not only fragment heavier nuclei into lighter ones but also produce species not in the primary cosmic ray flux (neutrons being of particular concern). The shielding "transport" computations are thus quite complex, involving nuclear physics models and a complex series of events in shielding material including secondary and tertiary interactions and finally absorption in tissue, and calculation of biological risk measures such as dose equivalent. The development and testing of such transport codes is not considered complete by the community involved in this subject, so NASA's space radiation shielding program-SRSP includes further development of these transport codes with a goal of producing a set of validated, documented and user-oriented methods for calculating the dose equivalent (or other measures of biological risk) within a complex spacecraft.

Polymers and composite materials that have mechanical and other properties suitable for "multi-purpose" applications are being developed under the SRSP to support space exploration. These materials need their shielding effectiveness in the GCR environment to be quantified and verified before including them in the vehicle design stage.

The methods of monitoring radiation exposure for flight crews are continuously being evaluated with comparisons between instruments and with calculations, and new dosimetry methods are being proposed. Radiation monitoring instruments needed on space exploration missions will evolve from current techniques and with future developments. The resulting dosimeters and spectrometers need to be tested in the GCR environment.

The deep space test bed (DSTB) addresses all of the above issues by exposing shielding materials and instruments to the full deep space cosmic ray energy spectra and elemental abundances. Comparing the measured fluxes of particles above and below shielding materials also gives a realistic target to test the radiation transport codes, since the data includes the effects of all the complexities that the codes must simulate. There are some differences between the free-space cosmic ray flux and that encountered at balloon altitudes near the poles which will be included in measurements and calculations.

The DSTB is a cost-effective way to meet two principal challenges for the SRSP; validating NASA's radiation transport codes and measuring the effectiveness of candidate shielding materials in the complex deep space cosmic ray flux that manned missions will encounter. The test bed will also be used to test, in a realistic environment, new instruments that may be used for deep space monitoring.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The DSTB Facility is an innovated means to access the interplanetary ionizing radiation without the necessity of launching a space flight payload. The facility uses sub-orbital flights to provide this access. These exposures include the full energy spectrum and composition of the galactic cosmic rays and are a cheap alternative to conducting exposures at particle accelerators to investigate many relevant affects of this natural occurring radiation.

FINAL REPORT
Element 1 Tasks

Develop Engineering Requirements: This task was completed during the first year. A workshop was convened for scientists interested in the objectives of the radiation shielding program. Input from these scientists concentrated on the types of investigations to be conducted on the DSTB. Their input was included in developing the design requirements for the DSTB. The workshop report is available at http://sd.msfc.nasa.gov/ A copy of the workshop proceedings and report are available on a CD by sending a request to the PI.

Design factors for the DSTB: The design details were completed during the second year of this effort. We developed a preliminary design that incorporated the input from potential users. The preliminary design was reviewed by an engineering team and distributed to the potential users for their comments. Forty four discrepancies, comments and suggestions were recorded from this review. A detailed design for each subsystem was developed that responded to the recommendations of the engineering team and user community. The resulting design completed a full technical review in April 2004 and was reviewed by the National Scientific Balloon Facility (NSBF). A copy of the detail design documents is available from the PI. Final certification of the DSTB design by NSBF was obtained after completing an engineering test flight in the USA in June 2005.

Radiation Environment: The GCR environment at balloon altitudes in the polar region was re-evaluated and updated to reflect the higher altitudes now attainable on polar long- duration balloon (LDB) flights. NSBF has recently increased their launch load capability that now allows the use of larger balloons resulting in higher float altitudes. The new GCR flux estimates are based on float altitudes of 125,000ft - 128,000 ft., which reduces the atmospheric overburden from 4.5 g/cm2 to 3.3 g/cm2. The CREME code was used to estimate the flux for hydrogen, helium, nitrogen and iron nuclei for these altitudes using the flight path of a previous balloon flight.

Measurements of the GCR flux have been made using passive detectors as piggy-back payloads on past balloon flights.

Task Progress:	These dosimeters were analyzed and compared with calculated LET spectrum and angular distribution of the incident particles. The detectors also measured dose and were compared with historical data in LEO and with the model we use to predict radiation exposures during a polar LDB flight. Influence of the DSTB on the radiation environment: Two simulation efforts were completed: a study of the background radiation levels at balloon altitudes and a preliminary assessment of the influence of the DSTB structure itself on the radiation environment. The background simulations used HETC and GEANT to evaluate the largest background signals, namely singly charged particles consisting of primary and secondary protons and other secondary particles such as electrons, pions as well as photons and neutrons. The results were generally comparable for a simple slab geometry of the gondola and atmosphere. The GEANT simulation was extended to include a 3-dimensional geometry for the gondola and atmosphere and provides a more realistic assessment of the expected background radiation at balloon altitudes. These results were compared with data acquired with the DSTB neutron monitoring system used during the test flight. Develop Operating Plans: A Draft Project Plan was developed for the DSTB. This included the operating plans for utilizing the DSTB and the interfaces with the user community and the NSBF that provides the flight operations. The Project Plan included Management Plans according to the relevant NASA manuals that govern information generating projects like the DSTB, WBS structure, Configuration Management, Schedule, Assembly Integration and Test plans, Verification Plan, Risk Assessment, Facility Management and Safety. These plans were developed by contacting organizations within NASA that represent these elements and consulting with them for advice and the relevant standards. Since this project is utilized as a facility the plan includes a payload process cycle for selecting, integrating and conducting field operations for a flight; interface requirements for investigators' payloads that cover mechanical, power and data subsystems. A cost plan for operations was developed that include costs for NSBF flight operations, DSTB team field operations, and DSTB refurbishment and expendables costs. Additional cost estimates were made for improvements in the radiation monitoring instrumentation. A flight requested was submitted to NSBF for inclusion in their program planning process for the DSTB test flight for June 2005. Integration and compatibility tests (Figure1) were completed in May 2005 and the test flight was successfully conducted on June 18, 2005 and lasted for 8 hours at float altitude. This test flight included payloads solicited through the Alabama Space Grant Consortium that reviewed applications and selected 4 university payloads which included participation of 13 university students in these experiments. The payload complement included proto-type DSTB instruments that characterize the radiation environment. Element 2 Tasks Instrument Design Simulations: The Charged Particle Spectrometer (CPS) is comprised of four integrated detector systems: tracking, time-of-flight (TOF), silicon matrix (Si-Mat) and Cherenkov. These systems work in concert to provide the raw data needed to interpret the charge and energy of cosmic rays with energies between 100 MeV/amu and 10 GeV/amu. Simulations were conducted to design these detector systems and to estimate the required exposure factors to meet the science objectives. Proto-typing Detector elements: Proto-typing and testing individual detector elements were completed. In the time-of-flight system we developed and tested different scintillating bars in conjunction with a linear multi-anode photomultiplier tube (MAPMT). The selected scintillators have a 0.5?0.5cm2 cross-section and are 30cm long. Six bars are bonded to the MAPMT faceplate, which forms one TOF detector, that measures the scintillation signal, provides 1-dimensional position measurement and records the time of occurrence. The CPS uses 10 of these TOF units mounted side-by-side to form one contiguous plane for the TOF. The CPS has two of these planes with a vertical separation of >30cm to make the TOF measurements. The TOF uses a custom board developed under this effort with both analog and digital (timing) channels. The timing is measured with a commercial 128 channel VME TDC. The complete TOF readout chain was tested using ground level muons. Two different Cherenkov radiators were used in the test flight configuration to confirm the design calculations. The high energy Cherenkov radiator was not flown during the test flight because of the short duration and resulting low flux of high energy high charged particles. The pixilated silicon matrix used in the CPS was completed. The data acquisition for this detector was based on a custom design that included an FPGA as the central control element. This device communicates directly to the CPS computer. Testing was completed using ground level muons prior to the flight. The tracking and Cherenkov detectors use a custom ADC board previously used on cosmic ray experiments. The CPU interface to this ADC board was converted from two custom logic control boards (used in the previous application) to a single FPGA board. The updated interface to the CPU has completed using laboratory testing. The CPS data acquisition system is based on a PC104 single board computer that uses a stack to integrate all the detector control boards. Construction and Exposure: The subsystems of the CPS were assembled and integrated test of the CPS was completed on the ground using cosmic ray induced muons. The software program that controls the detector components, acquires data and manages the data storage was completed. A pressure vessel was developed with a thin window to mount the CPS for balloon flight exposures. The CPS was integrated in the DSTB gondola. Power and communication were provided by the DSTB subsystems as designed. The ground support software program was developed and used to control the instrument during the flight from the DSTB ground stations. Data was acquired during the flight and preliminary data analysis confirmed the detector operated satisfactory during the 8 hour test flight.
Bibliography Type:	Description: (Last Updated: 10/05/2009)

Articles in Peer-reviewed Journals	Adams JH, Adcock L, Apple J, Christl M, Cleveland W, Cox M, Dietz K, Ferguson C, Fountain W, Bogdan G, Ghita, Kuznetsov E, Milton M, Myers J, O'Brien S, Seaquist J, Smith EA, Smith G, Warden L, Watts J. "Deep space test bed for radiation studies." Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, 2007 Aug 27;579(1):522-5. http://dx.doi.org/10.1016/j.nima.2007.04.114 , Aug-2007